

MOTION

$$v_y = u_y + gt$$

01

MOTION

$$s_y = u_y t + \frac{1}{2} g t^2$$

02

MOTION

$$v_y^2 = 2 g s_y + u_y^2$$

03

MOTION

$$v_x = u_x$$

04

MOTION

$$s_x = u_x t$$

05

MOTION

$$u_y = u \sin \theta$$

06

MOTION

$$u_x = u \cos \theta$$

07

MOTION

$$F_g = mg$$

08

MOTION

$$v = \frac{2\pi r}{T}$$

09

MOTION

$$a_c = \frac{v^2}{r}$$

10

MOTION

$$F_c = F_{net} = \frac{m v^2}{r}$$

11

MOTION

$$F_g = \frac{GMm}{r^2}$$

12

MOTION

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

13

MOTION

$$\frac{T_a^2}{r_a^3} = \frac{T_b^2}{r_b^3} = \frac{4\pi^2}{GM}$$

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ELECTROMAGNETISM

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} = \frac{kQq}{r^2}$$

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ELECTROMAGNETISM

$$E = \frac{F}{q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{kq}{r^2}$$

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ELECTROMAGNETISM

$$V = \frac{\Delta U}{q}$$

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ELECTROMAGNETISM

$$B = \frac{\mu_0 I}{2\pi r}$$

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MOTION · 03

Final vertical velocity (also $v_y^2 = u_y^2 + 2g s_y$)

v_y final vertical velocity
 u_y initial vertical velocity
 s_y vertical displacement
 g acceleration due to gravity

MOTION · 02

Final vertical displacement

s_y displacement in the vertical (y) direction
 u_y initial vertical velocity
 g acceleration due to gravity
 t time (s)

MOTION · 01

Final vertical velocity

v_y final velocity in the vertical (y) direction
 u_y starting velocity in the vertical (y) direction
 g acceleration due to gravity; near Earth's surface $g = -9.8 \text{ m s}^{-2}$

MOTION · 06

Vertical component of launch velocity (not in formula book)

u_y vertical component of initial velocity
 u launch speed
 θ launch angle above horizontal

MOTION · 05

Final horizontal displacement

s_x horizontal displacement, also called the 'horizontal range'
 u_x initial horizontal velocity
 t time (s)

MOTION · 04

Final horizontal velocity equals the initial horizontal velocity

v_x final velocity in the horizontal (x) direction
 u_x starting velocity in the horizontal (x) direction

MOTION · 09

Centripetal (orbital) velocity

v velocity of revolving object (m s^{-1})
 r radius of orbit (m)
 T period of revolution (s)

MOTION · 08

Gravitational force — the weight (F_W in Unit 2)

F_g gravitational force (weight) in newtons (N)
 m mass of object in the field in kilograms (kg)
 g gravitational field strength (N kg^{-1}) or accel. due to gravity (m s^{-2})

MOTION · 07

Horizontal component of launch velocity (not in formula book)

u_x horizontal component of initial velocity
 u launch speed
 θ launch angle above horizontal

MOTION · 12

Newton's universal law of gravitation

F_g gravitational force between two objects (N)
 G gravitational constant (see table)
 M, m masses of the two objects (kg)
 r radial distance between objects (m)

MOTION · 11

Centripetal force

F_c centripetal force in newtons (N), also F_{net}
 m mass of revolving object (kg)
 v velocity of revolving object (m s^{-1})
 r radius of orbit (m)

MOTION · 10

Centripetal acceleration

a_c centripetal acceleration (m s^{-2})
 v velocity of revolving object (m s^{-1})
 r radius of orbit (m)

ELECTROMAGNETISM · 15

Coulomb's law — force between two charges r apart

F force between the charged particles (N)
 $\frac{1}{4\pi\epsilon_0}$ Coulomb's constant k
 Q, q charge on each particle (C)
 r radial distance between charges (m)

MOTION · 14

Law of periods — Kepler's ratio T^2/r^3 is constant for one central star

T period of revolution about the central star (s)
 r radius of orbit (m)
 M mass of central star (kg)
 G gravitational constant
 a, b subscripts denoting the two bodies

MOTION · 13

Gravitational field strength, by force on a mass or by distance

g field strength (N kg^{-1} or m s^{-2})
 m mass of object in the field (kg)
 M mass of the astronomical body (kg)
 G gravitational constant
 r radial distance from centre of body (m)

ELECTROMAGNETISM · 18

Magnetic field about a straight current-carrying wire

B magnetic field strength in tesla (T)
 μ_0 magnetic constant (see table)
 I current in the wire (A)
 r radial distance to the wire (m)
 $\frac{\mu_0}{2\pi} = 2 \times 10^{-7} \text{ T m A}^{-1}$

ELECTROMAGNETISM · 17

Electric potential difference when work is done on/by a charge

V electric potential in volts (V)
 ΔU change in electric PE — work done (J)
 q charge on particle (C)

ELECTROMAGNETISM · 16

Electric field strength, by force on a charge or by distance

E field strength (N C^{-1})
 F force on a charged particle (N)
 q charge on particle (C)
 k Coulomb's constant $= 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
 r radial distance from the charge (m)

ELECTROMAGNETISM

$$B = \mu_0 n I$$

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ELECTROMAGNETISM

$$F = BIL \sin \theta$$

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ELECTROMAGNETISM

$$F = qvB \sin \theta$$

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ELECTROMAGNETISM

$$\phi = BA \cos \theta$$

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ELECTROMAGNETISM

$$\text{emf} = - \frac{N \Delta(BA \cos \theta)}{\Delta t}$$

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ELECTROMAGNETISM

$$\text{emf} = - N \frac{\Delta \phi}{\Delta t}$$

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ELECTROMAGNETISM

$$I_p V_p = I_s V_s$$

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ELECTROMAGNETISM

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

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SPECIAL RELATIVITY

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

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SPECIAL RELATIVITY

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

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SPECIAL RELATIVITY

$$p_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

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SPECIAL RELATIVITY

$$\Delta E = \Delta m c^2$$

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QUANTUM PHYSICS

$$\lambda_{\max} = \frac{b}{T}$$

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QUANTUM PHYSICS

$$E = hf = \frac{hc}{\lambda}$$

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QUANTUM PHYSICS

$$E_k = hf - W$$

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QUANTUM PHYSICS

$$\lambda = \frac{h}{p}$$

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QUANTUM PHYSICS

$$n\lambda = 2\pi r$$

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QUANTUM PHYSICS

$$mvr = \frac{nh}{2\pi}$$

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Force on a moving charged particle in a magnetic field

F force on the particle (N)
 q charge on particle (C)
 v velocity of particle (m s⁻¹)
 B magnetic field strength (T)
 θ angle between velocity and field (°)

Force on a current-carrying wire (use Fleming's left-hand rule)

F force on the wire (N)
 B magnetic field strength (T)
 I current in the wire (A)
 L length of wire in the field (m)
 θ angle between wire and field (°)

Magnetic field inside a solenoid

B magnetic field strength in tesla (T)
 μ_0 magnetic constant
 n turns per unit length (m⁻¹)
 I current (A)

Faraday's law of induction (2nd version)

emf electromotive force (V)
 N number of turns in the coil
 $\Delta\phi$ change in magnetic flux (Wb)
 Δt time elapsed (s)

Faraday's law of induction (1st version)

emf electromotive force (V)
 N number of turns of wire in the coil
 B magnetic field strength (T)
 $A_{\perp}$ loop area perpendicular to the field (m²)
 Δt time elapsed (s)

Magnetic flux through a loop

ϕ magnetic flux in weber (Wb)
 B magnetic field strength (T)
 A area of the loop (m²)
 θ angle between field and the loop's normal (°)

Time dilation

t dilated (relativistic) time interval
 t_0 proper time interval
 v speed of the frame relative to observer
 c speed of light in a vacuum

Ideal transformer — turns ratio

V_p , V_s voltages across primary / secondary (V)
 n_p number of turns on the primary coil
 n_s number of turns on the secondary coil

Transformer — energy conservation relationship

I_p current in the primary coil (A)
 I_s current in the secondary coil (A)
 V_p voltage across the primary coil (V)
 V_s voltage across the secondary coil (V)

Mass–energy equivalence

ΔE energy (J) for a given amount of mass
 Δm rest mass (kg) converted to energy
 c speed of light in a vacuum (see table)

Relativistic momentum

p_v relativistic momentum (kg m s⁻¹)
 m_0 rest mass (kg)
 v speed of the object
 c speed of light in a vacuum

Length contraction

L contracted length (observer moving relative to object)
 L_0 proper length (observer at rest to object)
 v relative speed
 c speed of light in a vacuum

Photoelectric effect — kinetic energy and work function

E_k max kinetic energy of ejected electrons (J)
 h Planck's constant
 f frequency of incident light (Hz)
 W work function of the electron (J)

Photon (blackbody) energy

E energy of a photon (J)
 h Planck's constant
 f frequency of the light (Hz)
 c speed of light in a vacuum
 λ wavelength of the light (m)

Wien's displacement law

λ_{max} wavelength of maximum intensity for blackbody radiation
 b Wien's displacement constant (see table)
 T absolute temperature of the body (K)

Quantised angular momentum of an electron in an atom

mvr angular momentum of the electron, symbol L (kg m² s⁻¹)
 m mass of the electron (kg)
 v velocity of the electron (m s⁻¹)
 n atomic energy level
 h Planck's constant

Standing-wave condition for an electron in a hydrogen atom

n atomic energy level
 λ electron (Bohr) standing-wave wavelength (m)
 r Bohr radius of the electron orbit (m)
 $2\pi r$ circumference of the electron orbit (m)

de Broglie / photon momentum

λ wavelength (m)
 h Planck's constant
 p momentum of the photon / particle (kg m s⁻¹)

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

$$W = hf_0$$

$$B = \frac{1}{3}(n_q - n_{\bar{q}})$$

$$B = n_b - n_{\bar{b}}$$

$$L = n_l - n_{\bar{l}}$$

Baryon number from quark content

- B baryon number
- n_q number of quarks
- $n_{\bar{q}}$ number of antiquarks

Work function and threshold frequency

- W work function of the electron (J)
- h Planck's constant
- f_0 threshold frequency (Hz)

Rydberg's equation for the hydrogen spectrum

- λ wavelength of the emitted light (m)
- R Rydberg's constant (see table)
- n_f final energy level
- n_i initial energy level

Lepton number

- L lepton number
- n_l number of leptons
- $n_{\bar{l}}$ number of antileptons

Baryon number from baryon count

- B baryon number
- n_b number of baryons
- $n_{\bar{b}}$ number of antibaryons